$\Gamma = \{\mathrm{diag}(2^j, 4^j) : j = 0, 1, 2\}$ over F_7

$\gamma_0 = \mathrm{diag}(1, 1)$	$\longrightarrow$	$\Phi(\gamma_0) = (2, 6)$
$\gamma_1 = \mathrm{diag}(2, 4)$	$\longrightarrow$	$\Phi(\gamma_1) = (6, 6)$
$\gamma_2 = \mathrm{diag}(4, 2)$	$\longrightarrow$	$\Phi(\gamma_2) = (6, 2)$

Trace alphabet $T = \{2, 6\}$

$|\Gamma| = 3 \leq |T|^2 = 4 \leq |T|^4 = 16$